

On the Topological Constraints of Multi-Messenger Q-LINK Architectures Extension 1

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Abstract

The Q-LINK architecture uses a single messenger qubit ($k = 1$) to mitigate VQA barren plateaus. We investigate scaling the messenger register to $k > 1$. By extending the Hilbert-Schmidt trace derivation, we prove that while multiple messengers offer a linear relative variance advantage, they incur a dominant inverse-exponential penalty via the expanded Haar integration volume. Alternative topologies (DLA fracturing or intra-register entanglement) fail to overcome this suppression. We conclude that $k = 1$ is the strict global optimum.

1 Extended Trace Algebra and the Haar Volume Penalty

Let $\mathcal{H}_m = (\mathbb{C}^2)^{\otimes k}$ be a register of k messenger qubits, extending the Hilbert space to dimension $d = 2^{n+k}$. The traceless cost operator $\tilde{V} = V \otimes I_m^{\otimes k}$ yields a squared Hilbert-Schmidt norm $Tr(\tilde{V}^2) = \frac{2^k N}{n}$, where $N = 2^n$.

Compared to a Vanilla circuit of $n + k$ total qubits, the asymptotic relative variance ratio at Haar saturation is:

$$\mathcal{R}_k(n) = \frac{Tr(\tilde{V}^2)}{Tr(V_{Vanilla, n+k}^2)} = 1 + \frac{k}{n} \quad (1)$$

Despite this linear relative advantage, absolute gradient variance dictates trainability. By the Master Formula bounds for $U(2^{n+k})$, absolute variance scales inversely with the squared dimension:

$$Var[\nabla \mathcal{L}] \propto \frac{Tr(\tilde{V}^2)}{d^2} = \frac{\frac{2^{n+k}}{n}}{4^{n+k}} = \frac{1}{n \cdot 2^{n+k}} \quad (2)$$

Therefore, every additional messenger qubit suppresses the absolute gradient magnitude by a factor of $1/2$. The linear trace gain is overpowered by the exponential $O(2^{-k})$ Haar volume penalty.

2 Topological Constraints

To circumvent this $O(2^{-k})$ penalty, one must restrict saturation of $SU(2^{n+k})$:

- **Non-Interacting Messengers:** Generates redundant interaction terms, failing to increase Dynamical Lie Algebra (DLA) expressibility while still paying the exponential volume penalty.
- **Interacting Messengers:** Intra-register entanglement thermalizes the subsystem, scrambling phase information and destroying the topological integrity of the residual connection.

3 Symmetry-Enforced DLA Fracturing

Restricting the DLA by enforcing S_k permutation symmetry confines the messenger register strictly to the completely symmetric subspace. By Schur-Weyl duality, this fractures the DLA and bounds the reachable manifold.

The symmetric subspace dimension is $d_{sym} = k + 1$, reducing the total dimension to $D_{sym} = N(k + 1)$. The restricted trace evaluates to $Tr_{sym}(\tilde{V}^2) = \frac{N(k+1)}{n}$. Applying the Master Formula over $U(D_{sym})$ yields:

$$Var_{k,sym}[\nabla\mathcal{L}] \approx \frac{Tr_{sym}(\tilde{V}^2)}{D_{sym}^2} = \frac{1}{nN(k+1)} \quad (3)$$

Relative to the optimal $k = 1$ baseline ($Var_{k=1} \approx \frac{1}{2nN}$), the variance scaling is:

$$\frac{Var_{k,sym}}{Var_{k=1}} = \frac{2}{k+1} \quad (4)$$

Theorem Conclusion: Enforcing S_k symmetry successfully downgrades the penalty from exponential ($O(2^{-k})$) to polynomial ($O(k^{-1})$). However, since $\frac{2}{k+1} \leq 1$ for all integer $k \geq 1$, absolute variance strictly decreases.

4 Conclusion

The Q-LINK architecture cannot be beneficially scaled to $k > 1$. Adding messengers mathematically guarantees either exponential or polynomial volume suppression. The $k = 1$ configuration is uniquely optimal, breaking trace symmetry without catastrophic dimensional expansion.